THE PORTRAIT OF THE BODY

AI + PERSONAL CHOICE

Which moment is worth keeping?

An AI-assisted film project that combines amateur tennis footage with what each player wants to express.

PROJECT SOURCE

Tongji University
College of Design and Innovation

Alibaba
Orange Lion Sports

Information Visualization 2026 / Three-person course project

COURSE RECOGNITION

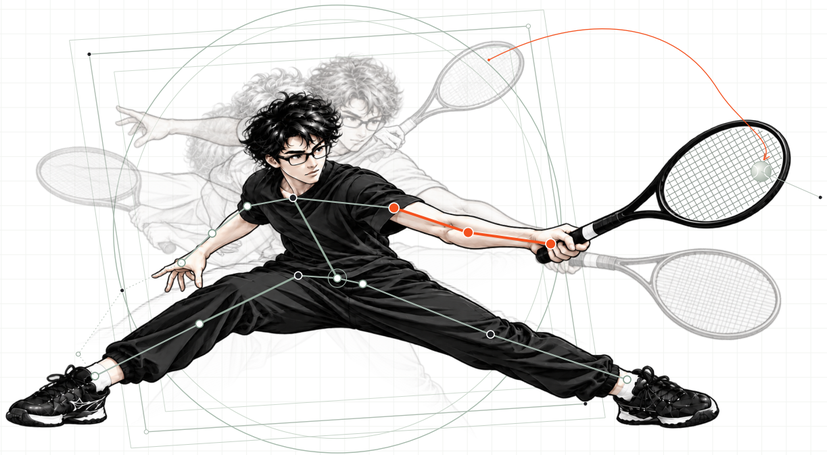
Best Creation Award

MY CONTRIBUTION

I proposed giving players a say in the film. I designed the choice system and portrait cards, implemented motion analysis and effects, and designed the exhibition interaction.

RESEARCH SCOPE

Four player interviews covered recording, revisiting, editing and sharing match footage.



AI-ASSISTED CONCEPT ILLUSTRATION

SELECTED FILM FRAME / COMIC

01 / Giorgia Lupi - Data Humanism

Put body data in the context of lived experience.

SELECTED FILM FRAME / LYRICAL

02 / Lin et al. - SportsBuddy, 2025

Link motion cues to creative editing decisions.

WHAT THE INTERVIEWS CHANGED

FOUR TENNIS PLAYERS

I01 / RECORDING TYPE

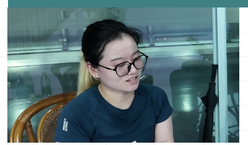
Keeps footage to revisit

They mostly rewatch clips in private and rarely share them. The footage is a record of time on court.

Design response: keep private archiving available.

E01 / 03:21–03:43 / LISTEN

I04 / REFLECTIVE TYPE



Wants a say in the edit

They know which rallies felt good. Automatic edits are quick, but the suggested highlights can miss what they would keep.

Design response: let the player confirm their own clips.

E03 / 02:21–02:44 / LISTEN

I03 / GROWTH TYPE

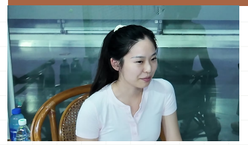
Wants to show how play feels

Asked about showing technique, they stressed the upbeat attitude they feel on court.

Design response: let the player set the film's tone.

E02 / 05:33–05:46 / LISTEN

I02 / EXPRESSIVE TYPE



Thinks twice before posting

They worry that showing a poor performance in a public post could invite criticism.

Design response: choose who sees the finished film.

I02 / 04:36–04:57

A rally can be worth keeping for different reasons.

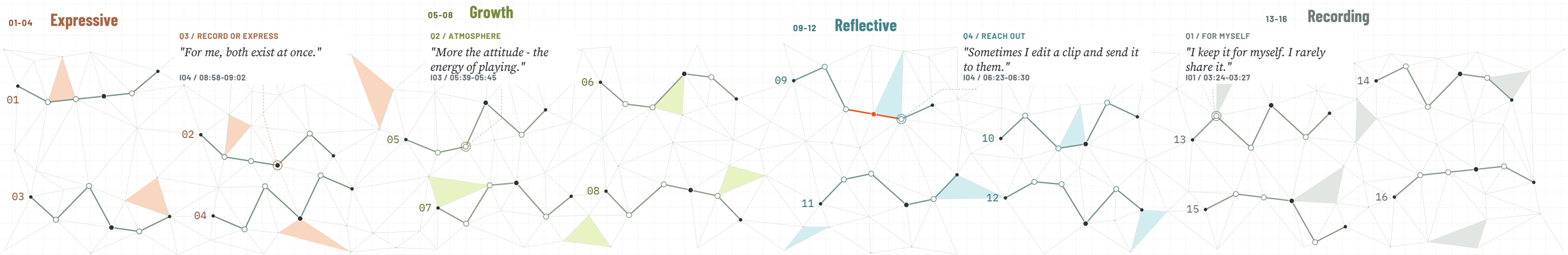
I proposed asking what the player wants this film to express before letting the system edit.

Paraphrased interview findings. The colours point to the four directions on page 02.

- The footage shows what happened. The player decides what matters.

Personal intent becomes an input.

I proposed bringing the player's intent into a workflow first planned as fully automatic, then designed four choices to do it.



16 ORIGINAL PATHS / ONE CONTINUOUS CONSTELLATION

Translated / edited interview excerpts inform the choice dimensions.

01 From interviews to design

	INTERVIEW SUMMARIES		DESIGN RESPONSE
I01 / E01	Keeps footage for private replay	→	Q1 / Ask the purpose and allow private saving
I03 / E02	Wants to show how play felt	→	Q2 / Ask whether technique or atmosphere leads
I04 / E03	Wants to choose the highlights	→	SELECT / Let players confirm clips or delegate

02 Four binary choices

Q1 + Q3 define groups. Q2 + Q4 vary them.

Q1 / MOTIVATION	For myself / For social media	Q2 / FOCUS	Technique / Atmosphere
Q3 / DATA USE	Record / Visual expression	Q4 / SOCIAL MODE	Reserved / Proactive

03 Path 13 / Recording

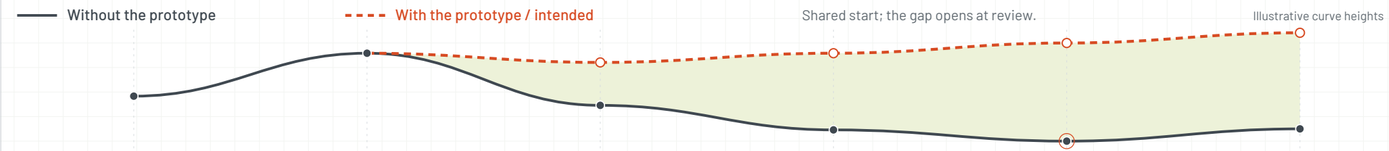
For myself / Atmosphere / Record / Proactive

CARD SUGGESTS — Jump cuts, light editing and ambient sound.

USER JOURNEY

From playing to a film worth keeping

Six stages / qualitative interview synthesis



	01 START PLAYING	02 A HIGHLIGHT	03 REVIEW FOOTAGE	04 TRY EDITING	05 SETTLE / STOP	06 SAVE / SHARE
WITHOUT / EFFORT AND UNCERTAINTY	Capture a tennis session.	Feel a rally is worth keeping.	Scan long clips; some rallies are missed.	Manual fixes interrupt the edit.	Settle for a template or abandon the edit.	Hesitate over public judgement.
WITH / CONFIRM, SHAPE, PREVIEW, CHOOSE	The same session.	The same moment.	AI suggests clips; the player confirms.	Set a tone and refine it with the crew.	Preview the film; revise the choices.	Save privately or choose an audience.

- Four choices set an initial direction. The prototype gives players a say in what follows.

Where the player takes over.

Six AI roles make the pipeline visible. Four player decisions shape what the film keeps, says and shares.

01 / SIX AI ROLES

READ THE MOVEMENT

01 Pose analyst
Tracks 33 body landmarks and supplies posture data.

02 Ball scout
Reads the ball and shot data for the player to check.

03 Highlight scout
Finds candidate moments and proposes a clip selection.

MAKE THE FILM

04 Scriptwriter
Proposes a storyboard, captions and an expression tone.

05 VFX artist
Applies the chosen treatment, from trails to impact effects.

06 Editor
Combines the shots, effects and music into the film.

THE CREW APPEARS IN ONE GROUP CHAT

Role status makes each handoff visible.

02 / PLAYER INTERACTION

ORIGINAL PROTOTYPE / SCRIPTED CHAT AND EXAMPLE VALUES

1 Verify the reading

Correct a reading before using it.

2 Choose the moments

KEEP OR DROP
Select the proposed clips that matter to you.

ADD A MOMENT
Flag a highlight the crew has missed.

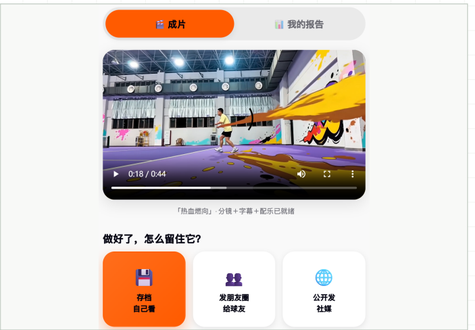
OR DELEGATE
Hand the selection back to the crew.

Keep, add or delegate the clip selection.

3 Set the tone

Confirm a tone or skip the decision.

4 Choose who sees it



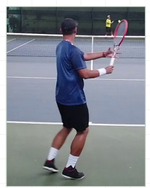
Preview, save privately or share.

03 / PRODUCTION PIPELINE

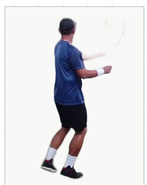
OUTLINED NUMBERS = AI ROLES

01 02 Pose analyst + Ball scout	03 Highlight scout	04 05 06 Scriptwriter / VFX / Editor	06 Editor / film + report
Pose and shot data	Candidate highlights	Parallel film production	Player decides the audience

04 / MY MOTION PIPELINE



SOURCE



PERSON CUTOUT

POSE

Same source frame at 00:10. The mask isolates the player. The landmarks capture the pose.

I implemented segmentation, pose analysis and timed effects.

05 / LIVE FOOTAGE TO AI COMIC

NO. 01 / SIMILAR FOREHAND POSES

LIVE FOOTAGE / 00:13.20

AI COMIC / 00:20.70

NO. 04 / MOTION / 00:01.50

Traces and position labels over the serve setup.

SAME-FRAME DETAIL

From a thread to a preview.

I made the choice system tangible. Follow a path, find its portrait card and try the treatment at the same table.

ONE VISITOR JOURNEY

01 WIND A PATH

Pull the thread from its hole. In each column, wind it around the post that matches how you want to express yourself.

Q1	Q2	Q3	Q4
FOR MYSELF	TECHNIQUE	RECORD	RESERVED
SOCIAL MEDIA	ATMOSPHERE	EXPRESS	PROACTIVE

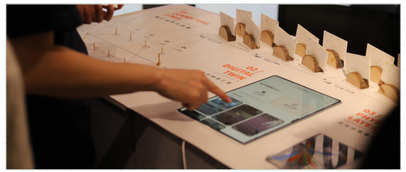
EXAMPLE ROUTE / CARD 13

→ 02 MATCH YOUR CARD

Match the line to one of 16 cards. Card 13 suggests jump cuts, light edits and ambient sound.

CARD 13 / ORIGINAL FRONT + BACK

→ 03 TRY THE PROTOTYPE



Continue on the inset iPad. Try the director crew, review the options and compare the example film with your intention.

→ 04 CHECK THE EFFECT

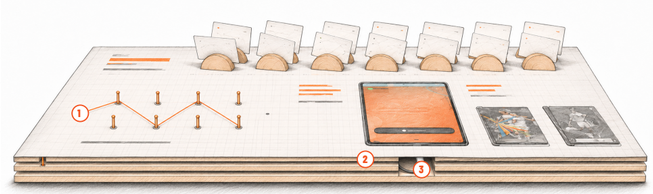
Stack or swap clear effects over a fixed Federer image. Two locating pins keep the layers aligned as you compare treatments.

→ Two kinds of feedback: a film example on screen and a visual layer you can swap by hand.

EXPLAINING THE STAND IN PERSON

I guided visitors through the project, connecting the poster and screen to the tabletop interaction.

BENEATH THE PRINTED SURFACE / TABLE DETAIL



01 / THREAD ON DEMAND

A small eyelet feeds the thread from below. Pull to begin.

02 / FIXED IN THE LAYERS

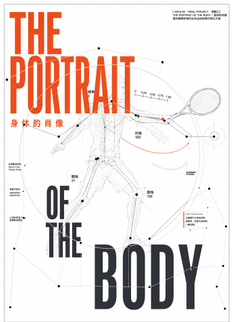
The wood seats the copper posts and recesses the iPad.

03 / CONCEALED POWER

The charging cable runs beneath the printed cloth.

EXHIBITION MEDIA

A KEY VISUAL POSTER



INSPIRED BY VITRUVIAN MAN

Inspired by Leonardo da Vinci's Vitruvian Man, the poster reframes the body through a circular frame, pose landmarks and swing trajectories. Condensed orange and black type turns movement into a graphic portrait.

- POSE LANDMARKS
- SWING TRAJECTORY
- Aa CONDENSED TYPE

B FIVE PROMOTIONAL FILMS

Each film gives a different player's movement its own visual treatment, from comic effects to animated typography.

ORIGINAL STILLS / FILMS 01, 02 AND 05

- The screen and transparent cards give visitors two ways to judge whether the treatment feels right.

A portrait, brought into public.

An exhibition that visitors could understand, choose, try and watch.

MY EXHIBITION WORK

I was responsible for the exhibition and interaction apparatus, bringing research panels, the film screen, winding area and portrait cards together as one stand. Visitors could understand the theme, choose, try and watch.

DESIGNING A PARTICIPATION SEQUENCE

01 **READ**
Find the project question.

02 **CHOOSE**
Wind a path; find a portrait.

03 **TRY**
Compare transparent effects.

04 **WATCH**
See the film possibilities.

TEAM RECOGNITION

AWARD PHOTOGRAPH / ORIGINAL EXHIBITION

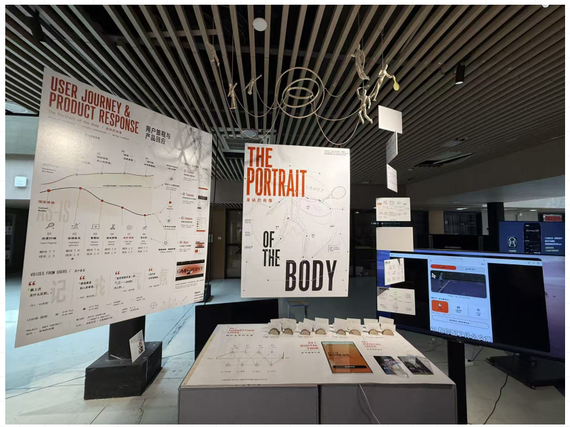
COURSE EXHIBITION / GROUP PHOTOGRAPH

ME

I helped bring personal ideas into generation, implemented underlying techniques and turned the choice system into a participatory physical experience.

This clarified my design interest: connecting personal expression, AI-assisted creation and interaction so technology responds to a particular person.

The project received the course collaboration's sole Best Award and recognition from Chengshi Sports.



FULL STAND / HANGING MODELS + TABLE

White surfaces, orange headings and black type extended the project's visual language. Hanging body models, lines and cards echoed the thread paths on the table, connecting the body, its traces and personal expression in one space.

Visitors could wind threads, find cards, stack transparent layers and watch the films. Five demonstration videos explored different directions, including live action, body contours and comic treatments.

5 film demonstrations / interaction prototype / physical apparatus / exhibition

- Connect personal expression, AI-assisted creation and interaction around a particular person.